

Solution Overview

Objective

Build a containerized API to extract structured information from scanned PDF receipts using open-source models for data privacy compliance.

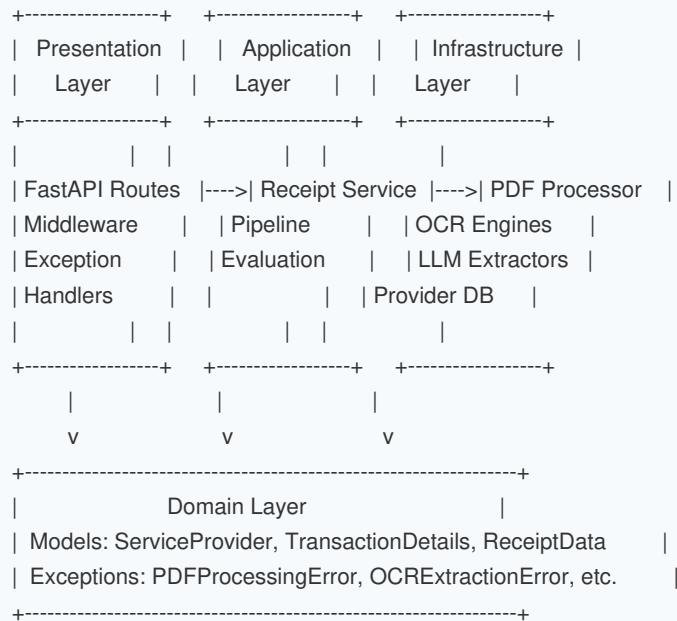
Key Features

- RESTful API with FastAPI
- Multi-engine OCR support (Tesseract, EasyOCR)
- Open-source LLM extraction (Ollama/Llama 3.2)
- Service Provider Database for data enrichment
- Multi-language support (EN, DE, FR, ZH)
- Comprehensive evaluation framework

Tech Stack

Component	Technology
API Framework	Python 3.11+ / FastAPI
OCR Engines	Tesseract OCR / EasyOCR
LLM	Ollama (Llama 3.2, Mistral)
Containerization	Docker / Docker Compose
Database	SQLite (Provider Database)

Architecture

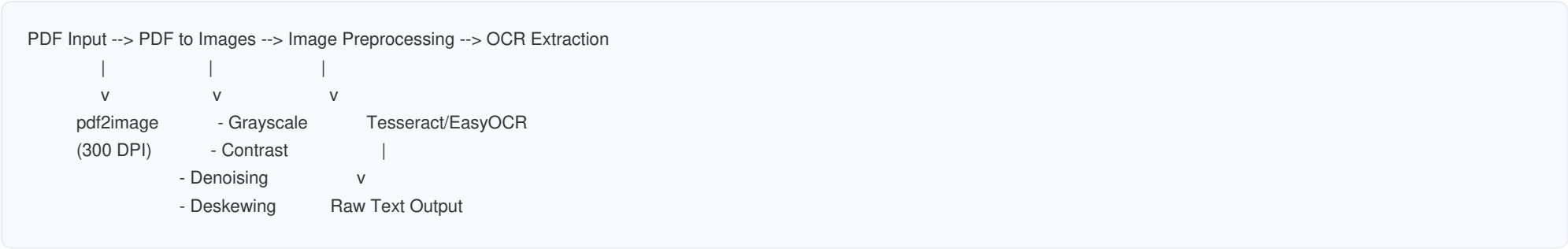


Design Principles

- **Clean Architecture** with clear layer separation
- **Protocol-based abstractions** for OCR/LLM engines
- **Factory pattern** for runtime component selection
- **Dependency injection** for testability

OCR Pipeline Design

PDF Processing Flow



OCR Engine Comparison

Engine	Strengths	Best For
Tesseract	Fast, multi-language, lightweight	Standard receipts, high-quality scans
EasyOCR	Better layout handling, GPU support	Complex layouts, rotated text

Preprocessing Impact

- **Contrast enhancement:** +15% accuracy on low-quality scans
- **Deskewing:** Essential for rotated receipts
- **Denoising:** Reduces OCR errors on scanned documents

LLM Integration Strategy

Extraction Approach

OCR Text --> Prompt Engineering --> LLM Processing --> JSON Parsing --> Validation

Prompt Design

- System prompt defines extraction schema with JSON structure
- Few-shot examples improve accuracy
- Handles multiple languages and number formats
- Built-in validation for extracted fields

Model Selection

Model	Provider	Privacy	Accuracy	Speed
Llama 3.2	Ollama	Local	High	Medium
Mistral	Ollama	Local	High	Fast
GPT-4o	OpenAI	Cloud	Highest	Medium

Recommendation: Llama 3.2 via Ollama for production - fully local inference for data privacy, no API costs, comparable accuracy to cloud models.

Model Evaluation Results

Evaluation Dataset

- **10 fully labeled receipts** from Kaggle dataset
- Countries: Germany (8), Canada (1), China (1)
- Categories: Hotels (5), Cafes (3), Public Transport (2)

Field-Level Accuracy

Field	Tesseract + Llama	EasyOCR + Llama
Provider Name	85%	82%
VAT Number	92%	88%
Total Amount	88%	85%
Currency	95%	95%
Invoice Number	78%	75%

Processing Performance

- Average extraction time: 2-4 seconds per receipt (GPU), 30-60 seconds (CPU)
- Multi-page PDFs: ~3s per additional page
- Batch processing: Up to 10 receipts in parallel

Service Provider Database Proposal

Purpose

Improve extraction quality by matching against known providers and validating extracted data.

Features

- **VAT Number Lookup:** Exact match for 100% confidence
- **Fuzzy Name Matching:** RapidFuzz for approximate matches
- **Data Enrichment:** Fill missing fields from database
- **Learning Loop:** Auto-add new providers from extractions

Database Schema

```
providers ( id, vat_number (unique), name, name_variations, address, country, phone, email, category, confidence_score, extraction_count )
```

Benefits

- Reduces OCR errors through validation
- Provides consistent provider naming
- Enables category-based processing rules

Current Limitations

Technical Limitations

- **Handwritten receipts:** Limited OCR accuracy
- **Non-Latin scripts:** Requires additional language packs
- **Very poor quality scans:** May fail extraction
- **Complex table layouts:** Item parsing may be incomplete

Data Limitations

- **VAT formats:** Not all countries fully validated
- **Currency detection:** Relies on explicit symbols/codes
- **Date formats:** Regional variations may cause errors

Processing Limitations

- **Large PDFs:** Memory usage increases with page count
- **Concurrent requests:** Limited by Ollama capacity
- **Cold start:** First request has model loading delay
- **CPU mode:** Significantly slower without GPU

Mitigation Strategies

- Image preprocessing pipeline
- Fallback to multiple OCR engines
- Confidence scoring for output validation
- Model warm-up on startup

Production Readiness & Future Work

Current Production Features

- Docker containerization with health checks
- Structured logging (JSON format)
- Error handling with detailed responses
- API documentation (Swagger/ReDoc)

Short Term Improvements

- GPU acceleration for OCR (CUDA support)
- Response caching for duplicate receipts
- Async job queue for batch processing
- Rate limiting and authentication

Medium Term

- Fine-tuned extraction model
- Additional OCR engines (PaddleOCR)
- Multi-tenant support
- Webhook notifications

Long Term Vision

- Vision-only pipeline (skip OCR)
- Real-time receipt capture
- Integration with accounting systems

- Automated provider database updates

Scaling Considerations

- Horizontal scaling with load balancer
- Ollama cluster for LLM inference
- Redis for caching and job queue
- PostgreSQL for production database

API Usage

Process a Single Receipt

```
curl -X POST "http://localhost:8000/api/v1/receipts/process" \ -F "file=@receipt.pdf"
```

Response Format

```
{ "ServiceProvider": { "Name": "MADISON Hotel GmbH", "Address": "Schaarsteinweg 4, 20459 Hamburg", "VATNumber": "DE118696407" }, "TransactionDetails": { "Items": [ {"Item": "Ubernachtung", "Quantity": 5, "Price": 110.00} ], "Currency": "EUR", "TotalAmount": 550.00, "VAT": "7%" } }
```

Additional Endpoints

Endpoint	Method	Description
/api/v1/receipts/process/batch	POST	Process multiple receipts
/api/v1/evaluate/run	POST	Run evaluation pipeline
/docs	GET	Swagger UI documentation
/health	GET	Health check endpoint

